MILITARY SPECIFICATION

DRILL, TWIST, BONE (Jacobs Chuck)

This specification is approved for use within DLA, Defense Personnel Support Center (DM), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Jacobs chuck bone twist drills suitable for bone drilling procedures.

1.2 Classification. Twist drills shall be of the following sizes as indicated, (see 6.1).

Size 1 - 1.1 mm diameter, 45 mm long

Size 2 - 1.5 mm diameter, 70 mm long

Size 3 - 2.0 mm diameter, 85 mm long

Size 4 - 2.7 mm diameter, 85 mm long

Size 5 - 3.5 mm diameter, 95 mm long

2. APPLICABLE DOCUMENTS

2.1 Government documents.

AMSC N/A FSC 6515

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER

23378 - Drill, Twist, Bone, Jacobs Chuck

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(Copies of specifications, standards and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92 Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The twist drill shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

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3.2 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier shall be responsible for preparing his own shop drawing. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure the proper function, fit, assembly or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Material. The drills shall be fabricated from type 440B corrosion-resisting steel.

3.4 Style, design and dimensions. The drill shall be of one-piece construction. The proximal end of the drill shall be round and non-notched for use in Jacobs chuck type hand and power drills. The drill shall conform to Defense Personnel Support Center (DPSC) Drawing No. 23378 for design and dimensions for applicable size number. The flutes of the drill shall be uniformly cut, shaped, and sharpened to cut at the tip as well as the sides of the operative end and provide automatic chip clearance. Size 1 (1.1 mm) drill shall be suitable for drilling holes for a 1.5 mm screw. Size 2 (1.5 mm) drill shall be suitable for drilling holes for a 1.5 mm cortex screw and thread holes for a 2.0 mm cortex screw. Size 3 (2.0 mm) drill shall be suitable for drilling gliding holes for a 2.0 mm cortex screw and thread holes for 2.7, 3.5 and 4.0 mm cortex screws. Size 4 (2.7 mm) drill shall be suitable for drilling gliding holes for 2.7 mm cortex screws. Size 5 (3.5 mm) shall be suitable for drilling gliding holes for 3.5 mm cortex screws.

3.5 Hardness. The drill shall be properly hardened, stress relieved and passivated. The drill shall have a Rockwell hardness of C56 to C58 when tested as specified in 4.4.2.

3.6 Runout. The total indicated runout of the drill shall not exceed 0.001 inch when measured as specified in 4.3.2.

3.7 Sterilization. In lieu of corrosion resistance requirement of GG-I-526, the drill shall be capable of withstanding repeated sterilization via steam, ethylene oxide, or long-term, 10 hour soaking in cold sterilization solution. The drill shall show no evidence of corrosion when autoclaved as specified in 4.4.3.

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3.8 Finish. The finish of the drill shall be smooth, bright, and uniform, except the operative ends shall have a fine honed finish with sharp cutting edges.

3.9 Marking. The use of a trademark is acceptable. The type of marking or placement of the marking shall not adversely affect or interfere with fit, function and operation of the drill with Jacob chuck type hand and power drills. The marking shall be distinguishable to the unaided eye.

3.10 Instructions. Adequate and accurate instructions shall be furnished with each unit on the use and care of the twist drills. Instructions shall include methods of sterilization and sanitization. The instructions shall also include the type of solutions, both suitable and unsuitable for cold sanitization of the drill.

3.11 Workmanship. The drill shall be free from defects which may detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of drills of the same size, manufactured under essentially the same conditions. Each unit package of drills shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

4.2 Quality conformance inspection. The examinations required to assure conformance of those drills to be offered for acceptance are classified as specified in tables II and IV (see 4.2.1 and 4.3) and 4.3.3. The tests required for acceptance are specified in table III of 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as shown in table II. The unit of product for examination purposes shall be the individual drill.

TABLE II. Examination criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.5
Major B	S-1	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table III. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be the individual drill, unless otherwise stated in the table.

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TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Inspection level
Chemical composition	3.3	4.4.1	See 4.2.3
Hardness	3.5	4.4.2	S-2
Autoclaving	3.7	4.4.3	S-1

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the drills shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same drills shall be used for the determination of two or more test characteristics.

4.3 Examination. The drills shall be examined for the possible defects listed in table IV. The examination shall not be limited to those defects shown in the table. Unless otherwise indicated in the table, the defect listed shall apply to all drills covered by this specification.

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TABLE IV. Classification of defects

Category	Defects
Major A:	
101	Drill size and design not as specified.
102	Drill not of one-piece construction.
103	Total indicated runout exceeds 0.001 inch.
104	Flute length not as specified for applicable drill.
105	Drill diameter not as specified for applicable drill.
Major B:	
151	Helix angle and Helix point angle not as specified.
152	Flute cross-section dimensions not as specified for applicable drill.
Minor:	
201	Drill not free from defects such as cracks or pits.
202	Finish on drill not as specified.
203	Overall length of drill not within specification limits.
204	Chamfer not as specified for applicable drills.
205	Angle for size 5 flat edge not as specified.
206	Instructions for use inaccurate, inadequate or missing.
207	Packaging and packing not as specified.
208	Marking on packaging or packing incorrect, illegible, incomplete or missing altogether.

4.3.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

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4.3.2 Runout. The drill shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the drill shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.3.3 Visual examination. Examination for possible drill defects not perceptible to the unaided eye shall be made using a 10X magnification.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2).

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

4.4.3 Autoclaving. The test drill shall be cleaned in accordance with the instructions supplied by the manufacturer. The drill shall then be autoclaved at 270 $\pm$ 5 degrees F. and 27 $\pm$ 1 psi for a minimum of 20 minutes. After the drill is removed from the autoclave, the entire surface area of the drill shall be examined for evidence of corrosion.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Drill shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each drill shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.1.4 Intermediate package. Twelve unit packages containing drills of one size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for drills shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, A or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units containing drills of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages (288 units) containing drills of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for drills shall conform to the MIL-STD-794 requirements for this level.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV and Type V.

5.2.5.1 Levels A and B. Drills packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Drills packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirement shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

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6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock Number	Item Identification	Size
6515-01-172-3229	DRILL, TWIST, BONE, 1.1 diameter, 45 mm long	1
6515-01-180-3231	DRILL, TWIST, BONE, 1.5 mm diameter, 70 mm long	2
6515-01-140-5189	DRILL, TWIST, BONE, 2.0 mm diameter, 85 mm long	3
6515-01-172-3228	DRILL, TWIST, BONE, 2.7 mm diameter, 85 mm long	4
6515-01-140-5190	DRILL, TWIST, BONE, 3.5 mm diameter, 95 mm long	5

Military Interests:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD-MB

Agent:

DLA-DM

Project No: 6515-3749

Location: NCR Disk 4

Civil Agency Coordinating Activities:

VA-OSS
PHS
FDA-MPQAS

06667

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-42031 (DM)	DRILL, TWIST, BONE (Jacobs Chuck)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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